WORKED SOLUTION

STEP I 2012, Question 1

Pure • Optimisation with Calculus

STEP I 2012, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

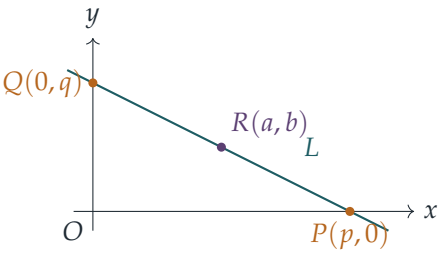
step.maths.org/questions/12-s1-q1

The key idea

Everything is controlled by one number, the gradient. Once L is forced through R, the intercepts p and q are simple functions of m, and both distances to be minimised become functions of m alone. The sum $p+q$ splits into a constant plus a term of the form $am + b/m$, which is a textbook job for AM–GM (or one derivative). For the length PQ it pays to notice that $q = mp$, which makes PQ^2 factorise nicely before we differentiate.

Solution

Finding p and q



Since $R(a,b)$ lies on $y = c - mx$, we have $b = c - ma$, so $c = b + ma$. The line meets the y-axis where $x = 0$, so $q = c$; it meets the x-axis where $y = 0$, so $p = c/m$ (allowed, since $m > 0$). Hence

$$p = a + \frac{b}{m}, \qquad q = b + ma.$$

Both are positive, as required, for every $m > 0$. Notice also that

$$q = b + ma = m\left(a + \frac{b}{m}\right) = mp,$$

which is just the statement that the gradient of L is $-q/p$. We shall use this below.

Minimising $OP + OQ$

Since $p, q > 0$, the distances of P and Q from O are p and q themselves, so as R stays fixed and m varies over $m > 0$ we must minimise

$$S(m) = p + q = a + b + \left(am + \frac{b}{m}\right).$$

The bracket is the sum of two positive numbers whose product, ab, does not depend on m. By the AM–GM inequality,

$$am + \frac{b}{m} \geq 2\sqrt{am \cdot \frac{b}{m}} = 2\sqrt{ab},$$

with equality exactly when $am = b/m$, that is, when $m = \sqrt{b/a}$ (which is positive, so this line is allowed). Therefore

$$S(m) \geq a + b + 2\sqrt{ab} = (a^{\frac{1}{2}} + b^{\frac{1}{2}})^2,$$

and the bound is attained. So the minimum value of the sum of the distances is $(a^{\frac{1}{2}} + b^{\frac{1}{2}})^2$, as required.

(If you prefer calculus: $S'(m) = a - b/m^2$, which vanishes only at $m = \sqrt{b/a}$, and substituting gives the same value.)

Minimising PQ

By Pythagoras, and using $q = mp$,

$$PQ^2 = p^2 + q^2 = p^2(1 + m^2) = \left(a + \frac{b}{m}\right)^2 (1 + m^2).$$

Minimising PQ is the same as minimising $f(m) = PQ^2$, since $PQ > 0$. Differentiating with the product and chain rules,

$$\begin{aligned} f'(m) &= 2\left(a + \frac{b}{m}\right)\left(-\frac{b}{m^2}\right)(1 + m^2) + 2m\left(a + \frac{b}{m}\right)^2 \\ &= 2\left(a + \frac{b}{m}\right)\left[-\frac{b}{m^2} - b + am + b\right] \\ &= 2\left(a + \frac{b}{m}\right)\left(am - \frac{b}{m^2}\right). \end{aligned}$$

Ask yourself Where does the $-b$ cancel?

Inside the square bracket, $-\frac{b}{m^2}(1 + m^2) = -\frac{b}{m^2} - b$, while $m(a + \frac{b}{m}) = am + b$; the two b's cancel. This cancellation is the reason for writing PQ^2 as $p^2(1 + m^2)$ rather than expanding $p^2 + q^2$ directly.

The first factor is positive, so $f'(m) = 0$ exactly when $am = b/m^2$, that is $m^3 = b/a$, giving the single stationary point

$$m = \left(\frac{b}{a}\right)^{\frac{1}{3}} = \frac{b^{\frac{1}{3}}}{a^{\frac{1}{3}}}.$$

(We are told we may assume it is a minimum; in fact $f' < 0$ for smaller m and $f' > 0$ for larger m, so it is the global minimum on $m > 0$.)

Now we substitute. With this m,

$$a + \frac{b}{m} = a + b \cdot \frac{a^{\frac{1}{3}}}{b^{\frac{1}{3}}} = a + a^{\frac{1}{3}} b^{\frac{2}{3}} = a^{\frac{1}{3}}(a^{\frac{2}{3}} + b^{\frac{2}{3}}), \qquad 1 + m^2 = 1 + \frac{b^{\frac{2}{3}}}{a^{\frac{2}{3}}} = \frac{a^{\frac{2}{3}} + b^{\frac{2}{3}}}{a^{\frac{2}{3}}}.$$

Hence

$$PQ^2 = a^{\frac{2}{3}}(a^{\frac{2}{3}} + b^{\frac{2}{3}})^2 \cdot \frac{a^{\frac{2}{3}} + b^{\frac{2}{3}}}{a^{\frac{2}{3}}} = (a^{\frac{2}{3}} + b^{\frac{2}{3}})^3,$$

and so the minimum distance between P and Q is

$$PQ_{\min} = (a^{\frac{2}{3}} + b^{\frac{2}{3}})^{\frac{3}{2}}.$$

This is the "similar form" asked for: the first answer has exponents $\frac{1}{2}$ inside and 2 outside, this one $\frac{2}{3}$ inside and $\frac{3}{2}$ outside.

Checks

- Symmetry: swapping a and b (reflecting in $y = x$) must leave both minima unchanged, and it does.
- A numerical case: take $a = 1$, $b = 8$. The second result predicts $m = 2$, giving $p = 5$, $q = 10$ and $PQ = \sqrt{125} = 5\sqrt{5} \approx 11.180$, and indeed $(1 + 4)^{3/2} = 5\sqrt{5}$. For the sum, $m = \sqrt{8}$ gives $p + q = 9 + 4\sqrt{2} \approx 14.657 = (1 + 2\sqrt{2})^2$.
- We checked both minima against a brute-force search over m for 2000 random pairs (a, b); they agreed to about 10^{-8}.

Common slips

- Expanding $p^2 + q^2$ in full and differentiating: correct but messy, and the factor $am - b/m^2$ is easy to miss. Spotting $q = mp$ first saves a lot of work.
- Forgetting that minimising PQ and PQ^2 are equivalent only because $PQ > 0$; and losing track of fractional powers when simplifying $a + b/m$.

Beyond STEP

The second result is the classic "ladder round a corner" problem in disguise: $(a^{2/3} + b^{2/3})^{3/2}$ is the longest ladder that can be carried horizontally round a corner between corridors of widths a and b. The family of segments PQ of fixed length envelopes an astroid, $x^{2/3} + y^{2/3} = \ell^{2/3}$, and R lies on the envelope exactly when ℓ is this minimum. Envelopes and constrained optimisation (later, Lagrange multipliers) appear in first-year calculus courses; Siklos, *Advanced Problems in Mathematics*, has more questions in this spirit.

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